

Performance Testing

Date	15 March 2026
Team ID	XXXXXX
Project Name	PocketSmart AI
Maximum Marks	5 Marks

The template requires testing approach, scenarios, results and evidence. filecite turn15file5L7-L16 No actual performance-test output was supplied, so the results are explicitly marked Not Executed rather than fabricated.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman for API response checks; JMeter/k6 can be used for concurrent-load testing.
Type of Testing	Planned: Load Testing, basic Stress Testing and API response-time testing.
Target Module / API	Recommendation generation endpoint; budget validation; recommendation comparison/refinement APIs.
Test Environment	Local development environment initially; hosted environment after deployment.
Test Date	To be scheduled after the end-to-end prototype is available.

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Normal recommendation request with valid budget and preferences	5	60	Successful responses with average response time below target.
2	Concurrent recommendation requests	20	120	System remains responsive and error rate remains below 1%.
3	Invalid / incomplete budget request	10	60	Validation errors returned without server failure.
4	AI/external API timeout or failure simulation	10	60	Graceful error response and no unhandled application crash.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	Not executed	Pending	Measure after deployed endpoint is available.
2	Response Time (Max)	< 5 seconds	Not executed	Pending	Measure under normal and peak load.
3	Throughput (Req/sec)	Record baseline	Not executed	Pending	Record successful requests per second.
4	Error Rate	< 1%	Not executed	Pending	Calculate failed requests / total requests.
5	CPU Utilization	< 80%	Not executed	Pending	Measure during load test.
6	Memory Utilization	< 80%	Not executed	Pending	Measure during load test.

Step 4: Observations & Analysis

No actual performance observations can be claimed from the supplied files. Execute the planned tests after the application and recommendation APIs are deployed, then update this section with measured values.

Step 5: Screenshots / Evidence

Pending. Attach Postman, JMeter or k6 results after testing.